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United States Patent

12412646

Kind Code

B2

Date of Patent

September 09, 2025

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Healthcare information analysis and graphical display presentation system

Abstract

Systems, methods, and computer-readable media for analyzing and presenting healthcare information are described. Some embodiments may include a system configured to receive healthcare information relating to a patient and to generate a patient profile. The patient profile may include a physiological status as well as a physiological assessment and a treatment assessment based on the automatic and dynamic analysis of the healthcare information. The healthcare information and the patient profile may be updated and/or accessed in real-time or substantially real-time through client logic devices in communication with the system. In this manner, a healthcare professional may enter healthcare information for a patient that is readily accessible by other healthcare professionals through the system. The system may present navigation objects that include a plurality of navigation layers selectively displayed based on user input. In addition, information objects may be displayed to users based on user navigation selections.

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Appl. No.: 17/686316

Filed: March 03, 2022

Prior Publication Data

Document Identifier

US 20220189594 A1

Publication Date

Jun. 16, 2022

Related U.S. Application Data

continuation parent-doc US 16833376 20200327 US 11282594 child-doc US 17686316
continuation parent-doc US 14800448 20150715 US 10636516 20200428 child-doc US 16833376
us-provisional-application US 62068518 20141024
us-provisional-application US 62024980 20140715

Publication Classification

Int. Cl.: **G16H10/60** (20180101); **G06F3/0482** (20130101); **G06F3/04842** (20220101);
G16H15/00 (20180101)

U.S. Cl.:

CPC **G16H10/60** (20180101); **G06F3/0482** (20130101); **G06F3/04842** (20130101);
G16H15/00 (20180101);

Field of Classification Search

CPC: G16H (40/63); G16H (15/00); G16H (20/30); G16H (80/00); G16H (30/20); G16H
(10/60); G16H (50/50); G06F (3/0482); G06F (3/011); G06F (3/04817)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2002/0055855	12/2001	Cule	705/2	G16H 15/00
2002/0170565	12/2001	Walker	128/920	G16H 15/00
2008/0098333	12/2007	Champion	715/849	G16H 50/70
2010/0050110	12/2009	Hughes	715/781	G16Z 99/00
2012/0041330	12/2011	Prichep	600/544	A61B 5/30
2013/0013342	12/2012	Morris	705/3	G16H 10/20
2014/0081655	12/2013	Schoenberg	705/2	G16H 40/67

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-9619774	12/1995	WO	G06F 19/325

OTHER PUBLICATIONS

A. Sarcevic, et al., "A paper-digital interface for information capture and display in time-critical medical work," 2012 6th International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth) and Workshops, San Diego, CA, USA, 2012, pp. 17-24 (Year: 2012). cited by examiner

Mohyuddin, W. A. Gray, H. Bailey, C. Jordan and D. Morrey, "Wireless Patient Information Provision and Sharing at the Point of Care using a Virtual Organization Framework in Clinical Work," 2008 Sixth Annual IEEE International Conference on Pervasive Computing and Communications (PerCom), Hong Kong (Year: 2008). cited by examiner

Z. Zhang et al., "The Five Ws for Information Visualization with Application to Healthcare

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority as a continuation application to U.S. patent application Ser. No. 16/833,376 filed on Mar. 27, 2020, which claims priority as a continuation application to U.S. patent application Ser. No. 14/800,448 filed on Jul. 15, 2015, which claims the benefit of U.S. Provisional Application Nos. 62/024,980 filed on Jul. 15, 2014 and 62/068,518 filed on Oct. 24, 2014, the contents of which are incorporated by reference in their entirety as if fully set forth herein.

BACKGROUND

(1) Advances in policies, assessment, and assurance aspects of injury prevention, pre-hospital care, acute care, and rehabilitation services have been driven by the systematic collection and analysis of injury data in mandated trauma registries. As front line providers of clinical care, trauma care professionals have ready access to and a profound understanding of such injury data. As such, trauma care professionals have a substantial role in the interpretation of these data to policy makers, the design and advocacy of injury control strategies, and the ultimate development of injury control policy.

(2) Nowhere is comprehensive, data-driven care more important than in the first six hours after injury or an acute illness (for example, stroke, acute coronary syndromes and severe sepsis). The outcome of a trauma or an acute illness is critically dependent on the provision of timely, evidence-based and complete care, for example, before hemorrhage and other forms of shock cause potentially irreversible organ injury or death. Such care often requires seamless coordination across disciplines and between multiple healthcare entities in order to focus the highest standards of care and the rapid and efficient deployment of resources in times of great crisis.

(3) Although trauma systems have made great strides in pre-hospital care and in the creation of highly functional trauma teams and trauma centers, there are still profound gaps in patients' access to uniformly high quality injury care and there are still great opportunities to improve safety and efficiency in the complex environment of acute trauma care. For instance, data generated at the point-of-care that could be used to inform complex decision-making or to improve health system performance is often not collected, lost, or not analyzed due to constraints in time or analytic power, and the chaotic environment of initial care. In addition, new developments in the understanding of the principles and practice of trauma care are often not accessible in a timely manner and are therefore not applied in circumstances when they might prevent a complication or save a life. Accordingly, patients would benefit from a system capable of collecting, analyzing, and presenting data generated at the point-of-care through timely and accurate techniques.

SUMMARY

(4) This disclosure is not limited to the particular systems, devices and methods described, as these may vary. The terminology used in the description is for the purpose of describing the particular versions or embodiments only, and is not intended to limit the scope.

(5) As used in this document, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the

art. Nothing in this disclosure is to be construed as an admission that the embodiments described in this disclosure are not entitled to antedate such disclosure by virtue of prior invention. As used in this document, the term “comprising” means “including, but not limited to.”

(6) In an embodiment, a healthcare information presentation system may include a client computing device comprising a processor and a non-transitory, computer-readable storage medium in operable communication with the processor. The computer-readable storage medium may include one or more programming instructions that, when executed, cause the processor to receive healthcare information from a server computing device in communication with the client computing device, present a navigation object on a display device in operable communication with the processor, the navigation object comprising a plurality of navigation levels including a primary navigation level and a secondary navigation level, each of the primary navigation level and the secondary navigation level comprising at least one selection area, receive a primary selection of a primary selection area of the primary navigation level, present at least one secondary selection area of the secondary navigation level based on the primary selection; receive a secondary selection of the at least one secondary selection area, and present at least one healthcare information object on the display device based on the secondary selection.

(7) In an embodiment, a computer-implemented method for presenting healthcare information may include, by a processor of a client computing device, receiving healthcare information from a server computing device in communication with the client computing device, presenting a navigation object on a display device of the client computing device, the navigation object comprising a plurality of navigation levels including a primary navigation level and a secondary navigation level, receiving a primary selection of a primary selection area of the primary navigation level, presenting at least one secondary selection area of the secondary navigation level based on the primary selection, receiving a secondary selection of the at least one secondary selection area, and presenting at least one healthcare information object on the display device based on the secondary selection.

(8) In one aspect, the primary navigation level and the secondary navigation level may be configured as concentric circles. In another aspect, the at least one healthcare information object may include a clinical practice guideline. In a further aspect, the at least one primary selection area may include at least two of an archive selection area, a support selection area, a team and roles selection area, a census selection area, an emergency department selection area, and a new patient selection area. In one aspect, the navigation object may be configured for a trauma event and the at least one primary selection area comprises at least two of a start selection area, a primary selection area, a secondary selection area, a flow sheet selection area, a review selection area, and a transfer selection area. In another aspect, the secondary selection area may be configured to access patient information for a trauma event and may include an arrival secondary selection area and a treatments on scene secondary selection area

(9) In an embodiment, a healthcare information analysis system may include a processor and a non-transitory, computer-readable storage medium in operable communication with the processor. The computer-readable storage medium may contain one or more programming instructions that, when executed, cause the processor to receive point-of-care healthcare information associated with a patient in substantially real-time collected via at least one of a plurality of mobile computing devices; analyze the healthcare information to generate a patient profile of the patient, the patient profile comprising a physiological status, a physiological assessment, and a treatment assessment, and generate at least one graphical user interface element associated with the patient profile for presentation at a plurality of display devices.

(10) In one aspect, the plurality of display devices may include a monitor device, such as an overhead monitor device. In one aspect, the healthcare information may include at least one of surgeries, symptoms, type of injury, severity of injury, mechanism of trauma, and trauma location. In an embodiment, the system may be configured to be used for trauma healthcare services and/or

surgical healthcare services. In an embodiment, the plurality of mobile computing devices comprise a smartphone and a tablet computing device. In one aspect, the healthcare information comprises user input and device input. In one aspect, the graphical user interface element comprises a dashboard, such as a dashboard configured to receive health information user input through at least one field. In one aspect, the dashboard is configured to present a graphical representation of a body of the patient for indicating an injury to the patient.

(11) In an embodiment, a computer-implemented method for analyzing and presenting health information may include, by a processor, receiving point-of-care healthcare information associated with a patient in substantially real-time collected via at least one of a plurality of mobile computing devices, analyzing the healthcare information to generate a patient profile of the patient, the patient profile comprising a physiological status, a physiological assessment, and a treatment assessment, and generating at least one graphical user interface element associated with the patient profile for presentation at a plurality of display devices.

(12) In an embodiment, a computer-readable storage medium having computer-readable program code configured to generate at least one healthcare assessment embodied therewith may include computer-readable program code configured to receive point-of-care healthcare information associated with a patient in substantially real-time collected via at least one of a plurality of mobile computing devices, computer-readable program code configured to analyze the healthcare information to generate a patient profile of the patient, the patient profile comprising a physiological status, a physiological assessment, and a treatment assessment, and computer-readable program code configured to generate at least one graphical user interface element associated with the patient profile for presentation at a plurality of display devices.

(13) In an embodiment, a graphical user interface for use in a healthcare environment to assist with treating patients may include a plurality of primary graphical objects, each of the plurality of primary graphical objects being associated with information regarding a patient, and a plurality of secondary graphical objects associated with the primary graphical objects such that selection of a primary graphical object effects presentation of at least one of the plurality of secondary graphical objects. In one aspect, the primary graphical objects may include a primary navigation level and the secondary graphical objects comprise a secondary navigation level. In another aspect, the primary graphical objects may be presented as an inner circle and the secondary graphical objects may be presented as an outer circle relative to the inner circle. In a further aspect, the graphical user interface may be configured for a trauma event and the plurality of primary graphical objects may include at least two of a start selection area, a primary selection area, a secondary selection area, a flow sheet selection area, a review selection area, and a transfer selection area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objects of the present invention will become more readily apparent from the following detailed description taken in connection with the accompanying drawings.

(2) FIG. 1 depicts an illustrative healthcare information analysis and presentation system according to a first embodiment

(3) FIG. 2 depicts a schematic diagram of a system according to some embodiments.

(4) FIG. 3 depicts an illustrative healthcare information analysis and presentation system according to some embodiments.

(5) FIG. 4 depicts an illustrative data flow for a healthcare information analysis and presentation system configured for trauma assessment within a healthcare facility according to some embodiments.

(6) FIGS. 5A and 5B depicts an illustrative data flow for a healthcare information analysis and

presentation system data integration according to some embodiments.

(7) FIGS. 6A and 6B depict illustrative security and privacy models for the health information analysis and presentation system according to some embodiments.

(8) FIG. 7 depicts an illustrative trauma system dashboard according to a first embodiment.

(9) FIG. 8 depicts an illustrative trauma system dashboard according to a second embodiment.

(10) FIG. 9 depicts an illustrative trauma system dashboard according to a third embodiment.

(11) FIG. 10 depicts an illustrative trauma system dashboard according to a fourth embodiment.

(12) FIG. 11A depicts an illustrative GUI template according to some embodiments.

(13) FIGS. 11B and 11C depict an illustrative template configured directed toward airway assessment according to some embodiments.

(14) FIGS. 11D and 11E depict an illustrative template directed toward trauma investigation according to some embodiments.

(15) FIG. 11F depicts an illustrative custom template configured to allow a user to enter vehicular accident trauma information according to some embodiments.

(16) FIG. 12A depicts an illustrative graphical user interface according to some embodiments.

(17) FIG. 12B depicts an illustrative overview dashboard according to some embodiments.

(18) FIG. 12C depicts a color scheme used within the healthcare information application according to some embodiments.

(19) FIG. 12D depicts primary and secondary categories according to some embodiments.

(20) FIG. 12E depicts an illustrative navigation bar according to some embodiments.

(21) FIG. 12F depicts an illustrative adjuncts bar according to some embodiments.

(22) FIG. 12G depicts a trend display according to some embodiments.

(23) FIGS. 13A-13C depict illustrative clinical practice guidelines (CPG) process displays according to a first embodiment.

(24) FIGS. 14A-14C depict illustrative clinical practice guidelines process displays according to a second embodiment.

(25) FIGS. 15A-15F depict various illustrative screens included in the health information application according to some embodiments.

(26) FIGS. 16A and 16B depict illustrative review category screens according to some embodiments.

(27) FIGS. 17A-17E depict illustrative start category screens according to some embodiments.

(28) FIG. 18A depicts an illustrative key patient indicator screen according to some embodiments.

(29) FIG. 18B depicts illustrative key patient indicators according to some embodiments.

(30) FIG. 19 depicts an illustrative screen depicting a clinical practice guidelines for a gunshot wound according to some embodiments.

(31) FIGS. 20A-20B depict an illustrative screen according to some embodiments for viewing and/or adding vitals and other patient information.

(32) FIG. 21 depicts an illustrative screen for accessing various portions of the body of a patient via a graphical representation thereof.

(33) FIG. 22 depicts an illustrative trauma representation screen according to some embodiments.

(34) FIGS. 23A-23E depict an illustrative GUI platform according to some embodiments.

(35) FIG. 24 depicts an illustrative screen template according to some embodiments.

(36) FIG. 25 depicts illustrative and non-limiting examples of technological advantages of a healthcare embodiment of the system.

(37) FIG. 26 depicts various symbols and icons that may be used within the healthcare information application to represent information and objects.

(38) FIG. 27 illustrates various embodiments of a computing device for implementing the various methods and processes described herein.

DETAILED DESCRIPTION

(39) The present disclosure generally relates to systems, methods and non-transitory computer-

readable media for collecting and analyzing healthcare information and generating and presenting healthcare assessments in real-time or substantially real-time. In particular, some embodiments provide a healthcare information analysis and presentation system (the “system”) that is configured to analyze, examine, search, investigate, consider, evaluate, and/or otherwise process healthcare information and to generate various physiological assessments and treatment assessments based on the healthcare information. In some embodiments, the system can generate graphical user interface (GUI) elements configured to present healthcare information, physiological assessments, and/or treatment assessments on a display device, such as a display device of a mobile computing device or display monitor in a manner that allows medical professionals to efficiently, effectively, and accurately provide healthcare to patients in a manner not available using conventional processes and technology.

(40) Healthcare information may generally include information associated with a patient receiving treatment through a healthcare entity. Non-limiting examples of healthcare information may include, without limitation, age, gender, weight, height, medications, surgeries and other medical procedures (for example, diagnostic tests, diagnostic imaging tests, or the like), occupation, past and current medical conditions, family history, patient description of health condition, healthcare professional description of health condition, symptoms, type of injury, severity of injury, mechanism of trauma, trauma location, healthcare professionals providing or assigned to provide care, or the like. A healthcare entity may generally include any entity capable of providing healthcare to a patient, including a hospital, a medical clinic, an outpatient facility, a doctor's office, a surgical center, a diagnostic facility, a medical specialist, an ambulance, an emergency room, a medical trauma team, a surgical team, or the like.

(41) In some embodiments, a patient profile of a patient generally includes information associated with the real-time or substantially real-time health status of a patient, for example, at the point-of-care by a healthcare entity. A patient profile may include information associated with physiological characteristics and treatment information of the patient. Illustrative and non-restrictive examples of information included in a patient profile may include patient physical characteristics and logistical information (e.g., height, weight, age, address, etc.), injuries, admission time, procedures performed and/or scheduled to be performed on the patient, diagnostic tests performed and/or scheduled to be performed on the patient, medical conditions, allergies, pregnancy status, patient medical status (e.g., “in shock,” hypothermic, conscious/unconscious, responsive, etc.).

(42) A physiological assessment may generally include any valuation, appraisal, evaluation, estimation, ranking, diagnosis, prognosis, and/or other calculation configured to indicate the physiological status of the patient based on the patient profile. For example, a physiological assessment may be generated indicating that a patient is likely experiencing a difficult airway condition based on information in the patient profile.

(43) A treatment assessment may generally include any valuation, appraisal, evaluation, estimation, ranking, and/or other calculation configured to determine a course of treatment for the patient based on the patient profile and the physiological assessment. Non-limiting examples of treatment assessments may include diagnostic testing, surgical procedures, medication, and any other type of treatment regimen for addressing medical issues indicated by the patient profile and/or the physiological assessment.

(44) The system configured according to some embodiments described herein provides multiple technological advantages and technical features. One non-limiting technological advantage and technical feature is the efficient capture of medical and patient data within standard processes of care, which may be analyzed in real-time or substantially real-time to provide effective and efficient point-of-care decision-making. Another non-limiting technological advantage and technical feature is the ability for all healthcare professionals involved in the assessment and/or treatment of a patient to document and retrieve medical and patient information in real-time or substantially real-time at the point-of-care on separate computing devices and/or display devices.

For example, a trauma nurse may update patient information from a mobile computing device in an emergency room and the updated patient information may be immediately available for access by a member of an operating team in the process of receiving the patient for surgery.

(45) A further technological advantage and technical feature is the ability to provide healthcare professionals with medical assessments and/or treatment determinations in real-time or substantially real-time at the point-of-care based on an analysis of information in the patient profile. For instance, the system may analyze the physiological information of a patient in view of historical medical data to determine a physiological status of the patient (e.g., cardiac arrest) and potential treatment regimens (e.g., medications, medical procedures). In this manner, the system is able to provide more effective and efficient medical evaluations and treatment recommendations to medical professionals compared to those available using existing processes and technologies (see FIG. 25 for illustrative and non-limiting examples of technological advantages of a healthcare embodiment of the system)

(46) The system provides multiple technological advances over and provides multiple technical features not present in traditional paper-based systems, conventional computer-based systems, and/or hybrid paper- and computer-based systems. Paper-based systems, such as conventional clinical charting techniques, are not capable of providing a user interface for interactive access to healthcare information, processes, or the like. In particular, traditional paper-based healthcare information systems rely on patient files with collections of charts and past medical records. Such patient files are not capable of being automatically or dynamically updated and do not provide access to a patient's complete medical history. Accordingly, healthcare professionals are not capable of accessing all of the information necessary to efficiently make accurate and reliable medical assessments using such paper-based medical files. In addition, healthcare professionals are not able to efficiently access the information that they need, as obtaining information requires physically searching through multiple documents, charts, and other files. Conventional computer-based systems suffer from much of the same deficiencies as paper-based systems, except that the healthcare provider is interacting with a computer screen instead of a paper file.

(47) Although a computer is able to locate and process information much faster, such conventional computer-based systems are not configured to present the information in an efficient, meaningful way that assists healthcare professionals with making faster and more accurate decisions for patient care. Conventional computer-based systems require healthcare professionals to go through myriad tedious drop-down selections, pages, and search queries in order to access information. Conventional computer-based systems are able to present information faster, however, they are not able to present meaningful information that assists healthcare professionals with efficiently sharing information and making quick and accurate decisions.

(48) In contrast, the methods and systems described according to some embodiments reduce the time and cognitive effort required for healthcare professionals to access, quantify, and assess healthcare information. For example, an emergency room physician is better able to make efficient and accurate decisions about treatment options for a trauma patient using the methods and systems described according to some embodiments in comparison to conventional healthcare information techniques. In addition, the methods and systems described according to some embodiments assist healthcare professionals with effectively and dynamically sharing information, for example, between departments, healthcare facilities, or the like in a meaningful way that leads to faster and better healthcare decision making. For example, methods and systems described according to some embodiments would allow a trauma surgeon preparing to operate on a car accident victim to quickly and intuitively access the accident and on-site treatment information with one GUI selection and then to access the diagnostic imaging results with a second GUI selection without having to search through multiple documents or pages and/or to ask a colleague for the information, as would be required using a conventional healthcare information system. In another example, clinicians at the point of care may have access to patient-specific, evidence based practice

guidelines and checklists. In a further example, trauma teams can review an overhead GUI interface to check on key physiological data and essential tasks during the course of treatment and resuscitation. In this manner, the system may streamline non-verbal communication by effectively displaying healthcare information, clinical practice guidelines, alerts, key patient indicators, process checklists, or the like. Such shared overhead or computing device graphical user interface projects may operate, among other things, to promote team cohesion and a shared mental mode among a disparate team of healthcare professionals treating a patient.

(49) A system according to the present teachings may be configured to transform healthcare information into a format that is easily accessible to medical professionals. For instance, the system may be configured to transform healthcare information into medical assessments and into objects, object values, and/or characteristics of objects displayed on a graphical user interface. In some embodiments, the system may be configured to transform information into color schemes configured to indicate process steps, stabilization of a patient, or the like. In this manner, information may be transformed into graphical interface objects and/or characteristics thereof that may be used to allow medical professionals to more efficiently, effectively, and accurately provide patient care, especially in time-sensitive trauma situations, than is possible using conventional techniques and processes.

(50) The system presents novel software tools and user interfaces that solve technical problems relating to providing medical care to patients, particularly in the real-time environment of trauma care. A non-limiting example of a technical problem that is solved by the system is providing efficient and effective access to all of the information necessary to treat a patient from a single point of access. Using conventional technology, such information is located in disparate locations, including paper charts and separate databases (e.g., vitals, demographic information, trauma event information, or the like). Thus, the use of such conventional technology can result in consuming valuable time to obtain the necessary information for treating a patient. For example, a physician in an emergency room may have to consult a paper chart or an electronic chart accessible through a computing device to obtain information concerning how the patient's injuries occurred. The treating physician may then have to consult another source to determine the patient's current vitals and yet another source to locate what medications and/or fluids, if any, the patient has received. The treating physician may then have to also consult with another source to determine which diagnostic tests have been completed and the results thereof. During this time, the treating physician may not have access to accurate information regarding how much time has elapsed since the trauma event or where the patient is in the treatment process.

(51) A system according to various embodiments of the present teachings solves these technical problems, as well as multiple others, by centralizing the information relating to the patient and any treatment thereof and presenting this information to medical professionals in a user friendly and efficient manner. The system also provides readily accessible timing information concerning the trauma event and/or treatment and where the patient is in the treatment process from a central access point. The system also solves the technical problem of allowing a user to efficiently navigate in an intuitive way through all of the information available within the system. As described below, patient information and treatment processes are accessible through easy-to-use, intuitive, and effective navigation tools and information presentation interfaces. In this manner, medical professionals are able to more completely, accurately, and efficiently access information required to treat patients. As such, the systems according to the present teachings provide a technological advantage over current techniques and technology.

(52) For example, in a trauma care setting, the system may be configured to: streamline the collection of clinical data at the point of care during trauma resuscitation and other acute clinical contexts to support relevant and complete documentation; link point-of-care data to other clinical data sources and resources for best practices; provide real-time data analytics to support clinical decision-making; enhance the communication of multidisciplinary health care teams; and create, in

real-time, deep data sets to inform safety, performance improvement, and research.

(53) As is well known in the art, the first 6 hours after severe injury or after the onset of other critical illnesses is a time period, where collection of data regarding the injury and making a decision how to treat the patient based on the collected data is of critical importance. The system may be configured for collection of standardized, high-resolution data by both physicians and nurses and to collect both point data from initial trauma surveys as well as minute-to-minute longitudinal data that can be used to display physiological trends. The system may be configured for the real-time analysis of both single point-in-time assessments and longitudinal data to provide more efficient and effective clinical assessments and to identify physiologic instability earlier during the course of illness. The system may include data warehouses and a big data analytics strategy that may provide regular and customized reports on quality of care and outcomes. The system may generate and maintain a deep data set, with significantly greater volume and detail than conventional trauma registries. As a result, the system can identify new predictive scores and previously unrecognized opportunities to improve patient safety and quality of care.

(54) FIG. 1 depicts an illustrative healthcare information analysis and presentation system according to a first embodiment. As shown in FIG. 1, the healthcare information analysis and presentation system (the “system”) **100** may include one or more server logic devices **110** (or server computing devices), which may generally include a processor, a non-transitory memory or other storage device for housing programming instructions, data or information regarding one or more applications, and other hardware, including, for example, the central processing unit (CPU) **2505**, read only memory (ROM) **2510**, random access memory (RAM) **2515**, communication ports **2540**, controller **2520**, and/or memory device **2525** depicted in FIG. 25 and described below in reference thereto.

(55) In some embodiments, the programming instructions may include a healthcare information analysis and presentation application (the “healthcare information application”) configured to, among other things, receive and analyze healthcare information and generate patient profiles and graphical user interface (GUI) elements associated with the patient profiles. The healthcare information application may be configured to receive, process, analyze, present, control, or otherwise manage healthcare information for various healthcare services, conditions, facilities, specialties, entities, providers, or the like. Although emergency room or “trauma” healthcare services are used as an example herein, embodiments are not so limited, as the system and healthcare information application may be used in connection with any healthcare services or facilities capable of operating according to some embodiments, including, without limitation, hospitals, outpatient facilities, surgical facilities (including emergency general surgery (EGS)), doctor's offices, medical specialists offices, diagnostic imaging centers, oncologist facilities, dental offices, nursing homes, or the like.

(56) The server logic devices **110** may be in operable communication with client logic devices **105** (or client computing devices), including, but not limited to, mobile computing devices, such as laptop computers, smartphones, personal digital assistants (PDAs), tablet computing devices, mobile medical equipment, wearable measurement devices, or any other mobile computing device now known or developed in the future. In some embodiments, the client logic devices may also include server computing devices, personal computers (PCs), kiosk computing devices, medical equipment, televisions, display monitors. The client logic devices **105** and the server logic devices **110** may communicate within the system using various communication and data transfer protocols, such as any of the various protocols known to those having ordinary skill in the art. Non-limiting examples of such protocols include Bluetooth, hypertext transfer protocol (HTTP), Ethernet, WiFi, Health Level 7 International (HL7), cellular communication protocols (e.g., 3G, 4G, LTE, etc.).

(57) In some embodiments, the healthcare information application may be accessible through various platforms, such as a client application, web-based application, over the Internet, and/or a mobile application (for example, a “mobile app” or “app”). According to some embodiments, the

healthcare information application may be configured to operate on each client logic device **105** and/or to operate on a server computing device accessible to logic devices over a network, such as the Internet. All or some of the files, data and/or processes used for analysis of healthcare information and/or the generation of patient profiles and associated GUI elements may be stored locally on each client logic device **105** and/or stored in a central location and accessible over a network.

(58) In some embodiments, one or more data stores **115** may be accessible by the client logic devices **105** and/or server logic devices **110**. The data stores **115** may include healthcare information, healthcare assessment processes, historical information, and/or the like. Non-limiting examples of data stores **115** may include healthcare information and management systems (HIMS), electronic medical record (EMR) systems, radiology information systems (RIS), picture archiving and communications system (PACS), medical registries, the National Trauma Data Bank (NTDB) (United States), the National Trauma Registry (NTR) (Canada), medical information repositories, or the like.

(59) Although the one or more data stores **115** are depicted as being separate from the logic devices **105**, **110**, embodiments are not so limited, as all or some of the one or more data stores may be stored in one or more of the logic devices.

(60) A healthcare professional may enter healthcare information (e.g., “clinical data” or “data”) into the system **100** using the healthcare information application through a client logic device **105**. The healthcare information may be entered at the point-of-care, for example, in an ambulance transporting the patient to a healthcare facility, in the emergency room of a hospital, or within a patient examination room of a private medical practice. The healthcare information may be available through the client logic devices **105** in real-time or substantially real time after being entered into the system **100**. For instance, the healthcare information application may analyze healthcare information entered by a first medical professional using a first client logic device **105** and generate a medical diagnosis and a treatment assessment that is stored in a storage device within the system **100**. A second medical professional may access the healthcare information, medical diagnosis, and/or treatment assessment using a second client logic device **105** in real-time or substantially real time after the healthcare information has been entered into the system **100** by the first medical professional.

(61) FIG. 2 depicts a schematic diagram of a system according to some embodiments. As shown in FIG. 2, a system **200** may include various core system components **205**. In some embodiments, the core system components **205** may include servers (e.g. servers **110** depicted in FIG. 1) and data storage devices configured to execute the healthcare information application and to receive and store healthcare information. The core system components **205** may also include client computing devices (e.g., client logic devices **105** depicted in FIG. 1) configured to execute or access the healthcare information application. The client computing devices may be configured to enter healthcare information into the system **200** through the healthcare information application. For instance, a client computing device may be a tablet computing device (e.g., iPad® manufactured by Apple Inc. of Cupertino, California, United States) executing a client version of the healthcare information application configured to present a data entry interface on a display component of the tablet computing device. A user may enter healthcare information using the data entry interface. In another instance, a user may access a healthcare information presentation interface generated by the healthcare information application to access and view healthcare information associated with one or more patients.

(62) The core components **205** may be in communication with healthcare entity computing systems **210**, such as a HIMS, an EMR system, medical devices and equipment, and computing devices. In some embodiments, at least a portion of the core components **205** may be configured to transmit/receive data (i.e., healthcare information) to/from the healthcare entity computing systems **210** through various protocols (e.g., Bluetooth, HTTP, Ethernet, WiFi, HL7, etc.) and interfaces

(e.g., application programming interfaces (APIs)). For instance, a core component **205** server may receive healthcare information from a healthcare entity computing systems **210** medical device or server in communication with a medical device. In one example, a hospital may use a blood pressure monitor configured to wirelessly communicate patient blood pressure readings to a server computing device of a central healthcare entity computing system **210**. The blood pressure readings may be transmitted as healthcare information to the core components **205**. In another example, a server computing device of the core components **205** may poll healthcare entity computing systems **210** seeking updated information for storage in a storage device of the core components **205**.

(63) The core components **205** may be in communication with various peripheral devices **215**, such as communication devices (e.g., hubs, routers, etc.), mobile computing devices, wearable or personal measurement devices (e.g., devices or sensors configured to measure various physiological characteristics of a user, such as heart rate, oxygen levels, temperature, etc.). The peripheral devices **215** may be configured to receive, generate, and/or transmit healthcare information to the core components **205**. For instance, a peripheral device **215** may be configured as a wearable heart rate monitor that may transmit heart rate information about a user to the core components **205**, such as a server computing device configured to store the heart rate information in a digital patient record.

(64) FIG. 3 depicts an illustrative healthcare information analysis and presentation system according to some embodiments. As shown in FIG. 3, a healthcare information analysis and presentation system (or system) **300** may include a computing device **305** having a processor **310** and system memory **315**. The computing device **205** may include any type of computing device, such as the client logic device **105** and server logic devices **110** described in reference to FIG. 1. The processor **310** may be configured to execute a healthcare information application **320**. The healthcare information application **320** may be configured to receive external data **370**, user input **372**, and/or device input **374**, for instance, through the processor **310** and/or as stored or cached as local healthcare information **325** in the system memory **315**.

(65) The external data **370** may include information from any data source accessible by the system **300**, including, without limitation a healthcare entity computing system, a HIMS, an EMR system, a RIS, a PACS, the NTDB, the NTR, and/or any other type of data store having healthcare information, a health information library and/or cloud, a third-party database, or the like.

(66) In some embodiments, the external information **220** may include any information associated with a patient, treatment, or a diagnostic test, including, without limitation, any information associated with the physical and/or mental condition of a patient, symptoms, medical history, medications, family history, diseases, illnesses, conditions, surgeries, medical procedures, medical diagnostic tests, vital signs, lab results, associated healthcare providers, demographic information, allergies, responses to treatment, responses to medication, health insurance information, medical claims, medical costs, diagnostic processes, healthcare protocols, or the like.

(67) The user input **372** may include data, such as healthcare information, entered into the system **300** by a user. For example, user input **372** may be provided by a user through a GUI interface of the healthcare information application **320** presented on a display component of a client logic device. The user input **372** may be received by the healthcare information application **320** and stored as healthcare information **325**.

(68) The device input **374** may include input generated by a device, such as medical devices or equipment (e.g., blood pressure device, heart rate sensor, body weight scale, thermometer, etc.), a peripheral device (e.g., a wearable measurement device or sensor), or the like. In some embodiments, a device configured to generate device input **374** may be in communication with the system **300** and/or the computing device **305**, streaming the device input **374** in real-time or substantially real-time.

(69) The healthcare information application **320** may include various modules, programs, applications, routines, functions, processes, or the like (“components”) to perform functions

according to some embodiments described herein. In some embodiments, the healthcare information application **320** may include a patient profile component **335**, a physiological assessment component **340**, a treatment assessment component **345**, and/or a GUI component **350**. (70) In some embodiments, the components **335-350** may be configured to access and/or receive the external data **370**, user input **372**, device input **374**, healthcare information **325**, and/or healthcare analysis process **330** as described according to some embodiments herein.

(71) The patient profile component **335** may be configured to generate a patient profile **380** using, among other things, the healthcare information **325**. The patient profile **380** may include admission information for the patient, such as time of admission, reason(s) for admission, treating facility, initial evaluation information, initial diagnosis, initial course of treatment, or the like. The patient profile **380** may also include demographic and medical history information concerning a patient, including, without limitation, age, height, weight, name, address, occupation, gender, medical conditions (e.g., diabetic, HIV+, allergies), pregnancy status, or the like. The patient profile **380** may include admission information for the patient, such as time of admission, reason(s) for admission, treating facility, initial evaluation information, initial diagnosis, initial course of treatment, or the like.

(72) The patient profile **380** may include a physiological status, a physiological assessment, and/or a treatment assessment associated with a patient. The physiological status may include the physical condition (e.g., “patient vitals”) of a patient based on the health information. The physiological status may be formed from various physiological elements or fields configured to provide information about the physical condition of a patient. For example, the physiological fields may include the temperature, blood pressure, heart rate, responsiveness, and/or the like. In another example, the physiological fields may include one or more injuries associated with the patient (e.g., laceration on face, burn on 45% of torso, tenderness on forearm, etc.) and the source or mechanism of the injuries (e.g., automobile accident, fall, etc.).

(73) The physiological assessment may include diagnoses of the patient by a healthcare professional and/or the generation of a diagnosis automatically and dynamically through the physiological assessment component **340** of the healthcare information application **320**. Illustrative physiological assessments may include, without limitation, a prognosis (e.g., predictive and prognosis scores), severity of injury scores (e.g., ISS), severity of illness scores (e.g., APACHE), transfusion requirements (e.g., ABC and TASH scores), clinical practice guidelines (CPGs), injury determination (e.g., burn, laceration, etc.), and/or a determination that a patient is experiencing shock, hypothermia, an allergic reaction, a difficult airway condition, cardiac arrest, or the like.

(74) In some embodiments, the physiological assessment component **340** may be configured to perform analytics on or to otherwise analyze the healthcare information **325** using the healthcare analysis process **330** to generate a physiological assessment. In some embodiments, the healthcare analysis process **330** may include rules, algorithms, processes, and other analytical mechanisms configured to diagnose a patient based on the patient profile **380** and/or external data in real-time or substantially real-time. For example, a patient may be admitted into a hospital emergency room and a healthcare professional may provide user input **372** to the healthcare information application **320** concerning the patient physical condition, injuries, or the like. Device input **374** may also be received by the healthcare information application **320** through medical devices and equipment configured to measure the patient's physical condition (e.g., “patient vitals”). The healthcare information application **320** may store the user input **372** and device input **374** as healthcare information. The patient profile component **335** may generate a patient profile **380** from the healthcare information. The physiological assessment component **340** may analyze the patient profile **380** using the healthcare analysis process **330**, including, without limitation, diagnostic algorithms (e.g., difficult airway algorithms, blunt force trauma algorithms, cardiac trauma algorithms), comparisons with historical data obtained through external data **370** (e.g., compare patient profile with healthcare information of other patients), injury prediction scores, prognostic

scores, or the like. The physiological assessment component **340** may analyze the patient profile **380** to generate trends associated with the healthcare information.

(75) The treatment assessment may include a treatment and/or diagnostic regimen or plan for the patient by a healthcare professional and/or automatically and dynamically through the treatment assessment component **345** of the healthcare information application **320**. For instance, the healthcare information application **320**, through the treatment assessment component **345**, may determine a treatment regimen for a diagnosed medical condition of the patient based on the patient profile. In some embodiments, the treatment assessment component **345** may be configured to perform analytics on or otherwise analyze the patient profile **380** using the healthcare analysis process **380** to generate a treatment assessment or plan for the patient. The healthcare analysis process **380** may include various processes, algorithms, decision trees, or the like to determine a course of treatment for a patient based on their physical condition and the diagnoses included in their physiological assessment. For example, the treatment assessment component **345** may determine that a patient requires a certain diagnosis test to determine the cause of a physical condition (e.g., abdominal pain), requires a massive blood transfusion, or is a candidate for a particular procedure (e.g., appendectomy).

(76) The GUI component **350** may be configured to provide GUI elements **382** that are graphical user interface elements and/or objects that can facilitate the entry of healthcare information **325** and can present to a user graphical representations of the patient profile **380** and any associated healthcare information, diagnosis, or treatment plan. The healthcare information application **320** may be configured to present the GUI elements **382** on a display component of a client computing device communicating with the system **300**. For example, a GUI element **382** may include a data entry interface for entering healthcare information associated with a patient, such as the patient's demographic and admissions information. In another example, a GUI element **382** may include a graphical representation of the progress of a patient through treatment, such as a trauma assessment and/or a surgical procedure. In a further example, a GUI element **382** may include a graphical representation of a patient's body with graphical indicators of medical conditions associated therewith. In a still further example, a GUI element **382** may include a patient dashboard displayed on a display device (e.g., television monitor, display monitor, etc.) in the healthcare facility, such as in an operating room, nurses' station, or waiting room. In some embodiments, a GUI element **382** may include a navigation object, for example, including a plurality of navigation levels.

(77) In some embodiments, the GUI component **350** may be configured to present a graphical representation of the body of the patient, for example, with indicators of injury, treatments, or the like arranged thereon. In some embodiments, the GUI component **350** may be configured to present the graphical representation of the body of the patient or portions thereof (e.g., an arm, a leg, etc.) responsive to the healthcare information application **320** receiving certain healthcare information. For example, the GUI component **350** may present an image of an arm responsive to healthcare information indicating injury to the arm such that a user may indicate and/or provide further information using a graphical representation of the body part. In this manner, a user of the system **300**, such as a healthcare professional, may interface with the system **300** using GUI interfaces and objects (e.g., data entry fields) to access and provide healthcare information. In some embodiments, the GUI component **350** may be configured to manage navigation of GUI screens and objects and to present GUI objects based on device input **374**, for example, such as presenting a particular secondary navigation level of a navigation object based on selection of an area on a primary navigation level.

(78) The system **300** may be configured as a mobile-device based platform designed for use by front line clinicians for the collection of data when providing healthcare services to patients. A non-limiting example of healthcare services include trauma assessment and resuscitation. In a trauma configuration, the system **300** may include modules that span trauma resuscitation in the first 6 hours (including nursing and physician documentation), the initial operation, the tertiary survey at

24 hours (a comprehensive assessment and final documentation of injuries), and the discharge summary (a module which summarizes a patient's course in hospital including complications and other outcomes). These key points of data capture characterize the main events in a trauma patient's treatment and recovery. The vast majority of data entered into the system **300** may be defined and standardized for ready integration into local and national trauma registries. Injury and other diagnostic fields, and key interventions may also be coded according applicable conventions, such as the Injury Severity Score (ISS) and the International Classification of Diseases (ICD) systems. (79) FIG. 4 depicts an illustrative data flow for a healthcare information analysis and presentation system (or system) configured for trauma assessment within a healthcare facility according to some embodiments. As shown in FIG. 4, the system **405** may be accessible to various departments **410a-g** of a healthcare facility, such as a hospital. For example, the system may be used within an outpatient department (OPD) **410a**, emergency medical services (EMS) **410b**, an emergency room (ER) **410c**, an operating room (OR) **410d**, an intensive care unit (ICU) **410e**, a medical ward (e.g., oncology ward) **410f**, or rehabilitation services ("rehab") **410g**. In some embodiments, the healthcare information **325** and patient profiles **380** associated with the system **405** may be updated in real-time or substantially in real-time and available in the healthcare facility departments **410a-g**. For instance, the physical condition of a patient (e.g., sudden loss of blood pressure) in the ER **410c** may be viewed by a healthcare professional in the OR **410d** as the patient is being moved to the OR **410d** for surgery.

(80) Data **415**, such as healthcare information **325**, healthcare analysis processes **330**, and/or patient profiles **382**, may be used for various functions **420a-i**. The system **405** may include a clinical documentation **420a** function through which medical professionals may document healthcare assessments (e.g., trauma assessments) and resuscitations simultaneously and in real-time on separate client computing devices, filling in non-overlapping and complementary data fields that may be accessed through multiple client computing devices and/or overhead monitors. The integrated documentation system may use a combined data set, created from trauma resuscitation records of multiple healthcare professionals, trauma registries, and other external data, in combination with healthcare information generated by the system **400** (e.g., physiological assessments and treatment assessments). In this manner, duplicate documentation efforts experienced in conventional systems may be reduced or even eliminated. The clinical documentation **420a** may be output data to printable resuscitation documents (i.e., reports) with reported fields and formats tailored to healthcare professional and/or legal documentation requirements.

(81) The system **405** may be configured to generate various alerts **420b** based on the patient profile **380** and associated healthcare information **325**. The alerts **420b** may facilitate point-of-care decision support, for instance, for trauma teams. In some embodiments, the healthcare information application **320**, for example, through the physiological assessment component **340**, may be configured to recognize highly critical situations based on the healthcare information **325**, the patient profile **380**, the user input **372**, and/or the device input **374** and to generate alerts **420b** responsive thereto. In some embodiments, the alerts **420b** may include heightened user prompts (e.g., GUI alert components displayed on a GUI interface) or clinical practice guidelines.

(82) Practice guidelines **420c**, such as clinical practice guidelines, injury-specific guidelines, and associated findings may be presented through a GUI interface on a display component of a client logic device. The practice guidelines **420c** may be determined based on the healthcare information **325** and patient profile **380**. For instance, the patient profile **380** may indicate (e.g., through analysis by the treatment assessment component **345**) that one or more clinical practice guidelines or injury-specific practice guidelines may provide appropriate treatment for a trauma patient's injuries.

(83) Checklists **420d** may be generated by the healthcare information application **320** responsive to specific clinical circumstances. In some embodiments, the checklists **420d** may be assembled

automatically by the healthcare information application **320** according to an individual constellation of injuries and presented on a GUI interface for review at critical phases in the trauma resuscitation. In some embodiments, the checklists **420d** may self-populate during the process of regular care and documentation as a checklist item is completed. In some embodiments, completion of checklist items may generate time-stamped documentation of completed items and detailed action during the process of trauma care. In general, the checklists **420d** are configured to be highly relevant to individual clinical circumstances, to limit their content to important and frequently omitted steps in clinical care, and to self-populate when tasks are accomplished during the regular processes of care. In some embodiments, the checklists **420d** may be configured to guide a medical professional team, such as a trauma team, through a systematic and complete approach to patient treatment. In some embodiments, checklists **420d** may be specific for a patient's condition.

(84) The healthcare information **320** and GUI elements **382** may be presented on one or more monitors **420e**, such as overhead monitors located at various locations throughout the hospital. The information displayed on the monitors **420e** may be updated by the healthcare information application **320** in real-time or substantially real-time to facilitate providing up-to-date and dynamic information in a dashboard format. The display of updated information through the monitors **420** may facilitate, among other things, team communication and optimization of trauma team responsiveness and use of resources.

(85) The system **400** may generate and utilize an electronic registry **420f** for analyzing healthcare information and diagnosing patients. In some embodiments, the electronic registry **420f** may be updated in real-time or substantially real-time and may include dynamic healthcare information of current patients as well as historical information from trauma registries (i.e., a “deep” registry including a greater volume and more detailed information than conventional registries or other medical databases). A “deep” electronic registry **420f** may facilitate identification of new predictive scores and previously unrecognized opportunities to improve patient safety and quality of care.

(86) The data **415** may be used by the healthcare information application **320** to provide performance improvement **420g** functions configured to improve the efficiency, efficacy, quality, and cost-effectiveness of care provided through a healthcare facility. In some embodiments, performance improvement **420g** may be implemented through benchmarking based on economic value of the healthcare services provided. In some embodiments, performance improvement **420g** may be at least partially based on the long-term outcomes of trauma patients, including their ability to return to work, reintegrate into society, and achieve high quality of life.

(87) The healthcare information application **320** may be configured to analyze the healthcare information **325** to provide mapping functions **420h**. For example, the mapping functions **420h** may generate maps for various medical conditions or traumas using healthcare information (e.g., de-personalized healthcare information). The maps, such as geographic information system (GIS) maps, may demonstrate patterns of medical conditions or traumas that may be used to further understand and even prevent certain medical conditions and/or traumas.

(88) As described above, the data **415** may be stored in various electronic registries **420f**, such as a trauma registry (e.g., NTDB, NTR, etc.). A research function **420i** may use these and similar registries and/or databases to perform research for various purposes, including improved patient diagnosis and treatment. In some embodiments, the research function **420i** may be used by the system **400** to allow for the improvement of the healthcare analysis processes **330**, the physiological assessment component **340**, and/or the treatment assessment component **345** (i.e., to “learn”).

(89) FIGS. 5A and 5B depict an illustrative data flow for a healthcare information analysis and presentation and data integration system according to some embodiments. As shown in FIG. 5A, the system data architecture **505** may include a flexible architecture that can be integrated with various existing platforms and processes, including third party systems. In some embodiments, the system data architecture **505** may process data using one or more translation engines **510**

configured, for example, to translate the health information and other data as stored within the system data architecture into formats capable of being integrated with other platforms, databases, and the like. The system may use two-way communication interfaces, such as the RESTful API or H7, to achieve data integration 515 to third party systems 520a-c while maintaining and adhering to strict security processes. This framework for the exchange, sharing and integration of patient data, can allow hospitals to realize the benefits of systems according to various embodiments of the present teachings as well as existing legacy information systems without major re-investment in new technologies. As shown in FIG. 5B, a single sign-on (SSO) authentication 525 allows for seamless authentication and authorization of users from a client computing device 530 for integration of data supporting IT in integrated system and role management, and providing users with one username and password to access various applications, such as EMR/HER 535 and/or system data and applications 540.

(90) FIGS. 6A and 6B depict illustrative security and privacy models for the health information analysis and presentation system (or system) according to some embodiments. The system may implement an integrated security and privacy model configured to support security and data protection at every single interaction point, including, for example, client logic devices within a healthcare facility 605, healthcare facility IT infrastructure 615, distributed computing systems (e.g., a cloud system) 610, and system IT infrastructure 620. The system may support security and HIPAA compliancy protocols known to those having ordinary skill in the art. Non-limiting examples of implemented security features within the system according to some embodiments include SSO and integrated user profiles, internal Wi-Fi authentication, mobile device management processes, on-device data encryption, firewall protection, data encryption in transit and at rest using specifications known to those having ordinary skill in the art. As shown in FIG. 6B, security functions may be implemented at various levels within the system, such as a system in which a client mobile computing device 625 operating at a healthcare facility (e.g., a hospital) may be in communication with the system 635 through a network, such as the Internet 630. For example, mobile device management 640a may allow for data wipes and updates of the client mobile computing device 625. User authentication 640b may be done at the healthcare facility, for example using SSO authentication and all data “at rest” may be encrypted 640e. In some embodiments, all data “in transit,” such as data being transmitted through the Internet 630 may be encrypted 640c, for example, through 256 bit encryption. The system 635 may implement a firewall 640d with managed access control and may encrypt 640e all data “at rest.”

(91) As described above, the system may generate GUI elements for receiving and/or presenting healthcare information and patient profiles. In some embodiments, the GUI elements may include user interfaces (or screens, windows, “dashboards,” or the like).

(92) FIG. 7 depicts an illustrative trauma system dashboard according to a first embodiment. As shown in FIG. 7, a trauma system dashboard 700 may include a trauma lifecycle element 705 configured to provide information associated with where the patient is in the trauma care timeline. The trauma lifecycle element 705 may indicate the duration of time since the patient was admitted and which steps have been completed, such as determining the mechanism of the trauma (M), determining injuries (I), survey (S), and transport (T). The dashboard 700 may include various patient profile information fields 710, 725, such as patient vitals and tests completed 710 and patient demographic information and medical conditions 725. In some embodiments, the dashboard 700 may present physiological and/or treatment assessments 715a, 715b and functions 730, such as alerts, protocols, and checklists automatically and dynamically generated by the healthcare information application 320. The dashboard may also present information associated with medical professionals and/or teams 720 that have been involved with the patient.

(93) FIG. 8 depicts an illustrative trauma system dashboard according to a second embodiment. As shown in FIG. 8, a trauma system dashboard 800 may include a patient profile navigation element 805 configured to allow users to access information objects configured to provide information

about the physiological status of the patient, including a primary navigation level **807a** and a secondary navigation level **807b**. A user may select a physiological status item or selection area, such as “pupils,” to cause information elements (status fields) **810** and other information objects associated with the item to be presented on the dashboard **800**. In FIG. **8**, selection of “pupils” on the patient profile element selection areas **805** has caused status fields **810** associated with the status of the patient's pupils to be presented on the dashboard. Healthcare information recorded in the database patient profile **815** may prompt the appearance of injury-specific clinical practice guidelines **820** on the dashboard. In addition, trends and historical data associated with the healthcare information depicted in the dashboard **800** may be graphically presented thereon. In some embodiments, the trauma system dashboard **800** may include a time elapsed information object **811** and/or timer **813** configured to indicate the time that has elapsed since an event, such as patient admission, trauma event, surgery, shock, or the like. In some embodiments, one or more of the selection areas **805** may be associated with a different time elapsed information object **811** and/or timer **813**. In some embodiments, the trauma system dashboard **800** may include a patient graphical representation selection object **817** that may allow a user to access a graphical representation of the patient, for example, such as depicted in FIGS. **10**,

(94) FIG. **9** depicts an illustrative trauma system dashboard according to a third embodiment. The dashboard **900** may be configured as an overhead dashboard (e.g., displayed on a common overhead monitor) to be viewed simultaneously by multiple members of a medical team, such as a trauma or surgical team. The dashboard **900** may include a graphical representation of the patient **905** which may include various medical conditions, injuries, or the like arranged thereon as well as a timeline **910** of the patient's medical conditions, diagnostic tests, procedures, or the like. The dashboard **900** may be configured to provide healthcare information associated with the volume of resuscitation fluids, including blood products, administered at a given point time **915a**, **915b**.

(95) FIG. **10** depicts an illustrative trauma system dashboard according to a fourth embodiment. The dashboard **1000** may include a graphical representation of the body of the patient **1005** configured to allow a medical professional to enter healthcare information associated with injuries to the patient. A list of injuries **1010** may be reported through the dashboard **1000**. For example, a user may select a portion of the body (e.g., the left arm) of the graphical representation of the body of the patient **1005**, e.g., by touching that portion, and may be presented with a menu of injury types and be prompted to select an injury type (e.g., laceration, burn, bleeding, sensitivity, skin discoloration, rash, abrasion, etc.) as well as information associated with the injury, such as the severity or degree of the injury. In other embodiments, rather than presenting a menu to the medical professional, the selection of an anatomical portion of the graphical representation of the body can allow the medical professional to type in an injury into a text field presented to the medical professional.

(96) In some embodiments, the GUI elements and interfaces may be configured based on GUI templates using a construction engine executed by the healthcare information application. A user may configure a GUI interface, such as a dashboard, using the construction engine. FIG. **11A** depicts an illustrative GUI template according to some embodiments. As shown in FIG. **11A**, a template may include tabs **1105**, blocks **1110a**, **1110b** and fields **1115a-c**. In some embodiments, each screen or dashboard rendered on a display component is a template **1100**. A template **1100** can contain one or more tabs **1105**, each tab **1105** can contain one or more blocks **1110a**, **1110b** of content, and each block **1110a**, **1110b** can contain one or more fields **1115a-c**. Each field **1115a-c** may provide a user with an option to select an associated value, for example, through typing a text or a number, making one or more selections, selecting an icon, etc. In some embodiments, selection of a value may cause the appearance of a window or other GUI object requesting more information and/or a function, such as a checklist, protocol guideline, or alert.

(97) Through the use of templates, users may configure dashboards using known GUI and data objects, such as fields. FIGS. **11B** and **11C** depict illustrative templates directed toward airway

assessment according to some embodiments. FIGS. 11D and 11E depict illustrative templates directed toward trauma investigation according to some embodiments. In some embodiments, a user may select various elements, such as an airway element **1120a**, a physiological criteria element **1120b**, or the like to include on a GUI. Although FIGS. 11A-11E depict a plurality of elements, only the airway element **1120a** and the physiological criteria element **1120b** are labeled to simplify the figures. A user may select an element **1120a**, **1120b** and link it to data and/or selection of a category. For instance, the physiological criteria element **1120b** may be linked to the systolic blood pressure of a patient and may be displayed when a particular secondary selection area (or “category”) **1212** is selected on the navigation object **1204**. In this manner, users may efficiently and effectively generate customized GUI interfaces, dashboards, and the like.

(98) FIG. 11F depicts an illustrative custom template configured to allow a user to enter vehicular accident trauma information according to some embodiments. For example, the template may include elements for specifying the type of vehicle **1125a**, the position of the vehicle/impact type **1125b**, whether the accident involved a rollover **1125c**, the number of vehicles involved in the accident **1125d**, speed of the vehicle at the time of the accident **1125e**, or the like.

(99) FIG. 12A depicts an illustrative graphical user interface according to some embodiments. As shown in FIG. 12A, a GUI may be presented on a display device of a client computing device **1205**. For instance, a GUI may be displayed on a touch screen of a tablet computing device. The GUI may include a navigation object **1215** that includes a plurality of navigation levels, such as primary graphical objects (for example, a primary navigation level) **1225** and secondary graphical objects (for example, a secondary navigation level **1235**). The navigation level is not limited to two navigation levels, as any number of navigation levels capable of operating according to some embodiments is contemplated herein. For example, the navigation object **1215** may include one, two, three, four, five, ten, fifteen, twenty, or any range of navigation levels between any two of these values. Each navigation level may include one or more selection areas **1245a-n**, **1255a-n** that may be selected by a user. In some embodiments, the presentation, configuration, and/or selection areas **1245a-n**, **1255a-n** of a navigation level **1225**, **1235** may be specified by the system based on the state of a navigation level and/or the selection of a selection area. For example, selection by a user of a selection area **1245a-n** on the primary navigation level **1225** may cause the presentation of the secondary navigation level **1235** and/or the display of particular selection areas **1255a-n** on the secondary navigation level. In another example, selection of selection area **1245a** may cause a particular secondary navigation level **1235** to be displayed that includes selection areas **1255a** and **1255c**. In a further example, selection of selection area **1245n** may cause a particular secondary navigation level **1235** to be displayed that includes selection areas **1255b** and **1255n**.

(100) The GUI depicted in FIG. 12A may include an information object **1265** that may include one or more information elements **1275a-n**. The information object **1265** may be configured to display information, such as healthcare information received from a server computing device. In some embodiments, the information object **1265** may include one or more information elements **1275a-n** configured to display data, such as data associated with a patient. The visibility of the information object **1265** and/or the particular displayed information elements **1275** may be determined based on the selected or active selection areas **1245a-n**, **1255a-n**. For instance, the selection by a user of selection area **1245a** on the primary navigation level **1225** may cause a particular secondary navigation level **1235**, with a particular set of selection areas **1255a-n**, to be displayed on the GUI. For example, the primary navigation level **1225** may include a diagnostics selection area **1245a-n** that, when active, may cause a diagnostics secondary navigation level **1235** to be displayed with selection areas **1255a-n** for particular diagnostic tests.

(101) Selection of a diagnostic test via the selection areas **1255a-n** may cause an information object **1265** to be presented on the GUI that has information elements **1275a-n** for displaying information associated with a particular diagnostic test for a subject patient. In some embodiments, portions of the information object **1265**, including the information elements **1275a-n** may be selectable by a

user and, for example, selection thereof may cause the presentation of other GUI elements, including, without limitation, information objects, information elements, navigation objects, and/or navigation levels. In some embodiments, the information object **1265** or a portion thereof may include a flow chart, clinical practice guideline, or other treatment process. The use of multiple navigation levels and information objects, such as the primary navigation level **1225** and the secondary navigation level **1235**, may allow a user to efficiently access and/or edit the vast amount information within the system in a minimal number of steps. For instance, in reference to FIG. **17A**, selection of the “1^o” selection area **1245** followed by selection of the “Start” selection area **1255** may cause the presentation of the “Treatments on Scene” information object **1265**, which may allow full access to the information elements **1275** associated with treatments given to the patient at the scene of a trauma.

(102) Although FIG. **12A** and certain embodiments provided herein describe the primary graphical objects **1225** and the secondary graphical objects **1235** as navigation objects or levels, embodiments are not so limited as the primary graphical objects and the secondary graphical objects may include any GUI object capable of operating according to some embodiments. In some embodiments, the selection of a primary graphical object **1225** effects the presentation of, the data associated with, and/or the event associated with a selection of the one or more of the secondary graphical objects **1235**. For example, selection of a first primary graphical object **1225** may cause a first secondary graphical object **1235** to be presented on the screen. Selection of the first secondary graphical object **1235** may cause a first selection event (for instance, navigation to a particular screen, data entry event, or the like). Selection of a second primary graphical object **1225** may cause a second and a third secondary graphical object **1235** to be presented on the screen. Selection of the second secondary graphical object **1235** may cause a second selection event and selection of the third secondary graphical object may cause a third selection events. In some embodiments, the primary graphical objects may be and/or may include selection areas **1245a-n** and the secondary graphical objects **1235** may be and/or may include selection areas **1255a-n**.

(103) The navigation object **1215** may be configured in various shapes and/or forms, including a circle shape, a rectangular shape, a square shape, a menu form, and/or any other shape or form capable of operating according to some embodiments described herein. As shown in FIG. **12B**, FIGS. **15A-15D**, FIG. **16A**, FIGS. **17A-17E**, and FIGS. **23A-23E**, the navigation object **1215** may have a circular or substantially circular shape. In some embodiments, the secondary navigation level **1235** may encompass, envelop, or otherwise surround the primary navigation level **1225**, such as through the primary and secondary navigation levels being configured as concentric circles, rectangles, or other concentric shapes.

(104) FIG. **12B** depicts an illustrative overview dashboard according to some embodiments. As shown in FIG. **12B**, an overview dashboard may include various GUI objects, such as a home bar **1202** (or notifications bar), a navigation object **1204** (or circle of life object or navigation wheel), and data entry objects **1206**. In some embodiments, the navigation object **1204** may be launched responsive to an event, such as logging into the system, accessing a patient record, or the like. In some embodiments, the notifications bar **1202** may be configured to provide information relating to a subject patient associated with the dashboard and data associated therewith, such as information relating to the patient's name, physical condition (i.e. “pregnant female,” and “age 32”), and source of trauma (e.g., “Motor Vehicle Crash”). In some embodiments, any alerts, messages, or other notifications may be presented on the notifications bar **1202**, such as treatment instructions, vitals outside of threshold values, messages (e.g., “transport patient to OR”), or the like.

(105) In some embodiments, the navigation object **1204** may include a timer **1208** configured to present a time associated with a procedure, patient, process, event, or the like. In some embodiments, the timer **1208** may be associated with a primary category **1210** and/or a secondary selection area **1212**. Time has a major effect on the outcome of patient treatment, particularly in trauma situations. Accordingly, the timer **1208** may operate to provide a constant reminder of time

passed since the occurrence of an event, such as a trauma suffered by a subject patient. In a trauma situation, stabilization within the first six hours increases the chance of long-term survival of the patient. As such, the timer **1208** may operate to provide medical professionals with a constant update on the time associated with treating the subject patient to allow the treatment team, among other things, to focus on time-based treatment goals. In some embodiments, multiple timers or timed events may be maintained within the system. Selection of a selection area **1210**, **1212** may cause a corresponding timer **1208** to be displayed. In some embodiments, there is one main timer maintained within the system, for example, time elapsed since admission, time elapsed since trauma, or the like which the timer **1208** may display.

(106) The data entry objects **1206** may allow a user to access and view health information relating to the patient, such as vitals, medications prescribed and/or taken, diagnostic tests, and trends thereof, or the like. For example, the data entry objects **1206** may include a selection object to view x-ray diagnostic tests relating to the subject patient.

(107) FIG. **12C** depicts a color scheme used within the healthcare information application according to some embodiments. As shown in FIG. **12C**, colors and color schemes may be used to identify information, status, processes, categories, or the like. In this manner, medical professionals may be able to readily identify information associated with a patient, process, category, or the like through the use of colors. As such, through training and/or familiarity with the healthcare information application, medical professionals can make associations between displayed colors and information and/or the treatment of a patient, which may be more efficient and effective than viewing the same information through text, menus, or other means of communication. For instance, the colors may parallel the stabilization of a patient in a trauma situation. In another instance, the use of colors and their relationship with data within the healthcare information application may operate to connect the role of data within the healthcare information application and what is actually occurring in the real world. In some embodiments, colors may be associated with the primary categories **1210**. For instance, “Start” may be associated with red **1214a**, “1°” (primary survey, for example, airway, breathing, exposure, circulation, and disability assessments) may be associated with orange **1214b**, “2°” (secondary survey) may be associated with yellow **1214c**, “Flowsheet” may be associated with pink **1214d**, “Review” may be associated with purple **1214e**, and “Transfer” may be associated with blue **1214f**.

(108) As shown in FIG. **12D**, the circular navigation object **1204** allows for intuitive, process-centric navigation. In some embodiments, the navigation object **1204** may include primary selection areas (or “categories”) **1210** (for instance, “Primary #2,” “Primary #3,” etc.) that may be selected and/or ordered based on standard ordering or procedures and processes within a particular medical unit. In the embodiment depicted in FIG. **12D**, the Primary #1 primary selection area **1210** has been selected and is the “Selected Primary” selection area. In some embodiments, the secondary selection areas **1212** may be presented based on the “Selected Primary” selection area **1210**. For example, in a trauma application, primary selection areas **1210** may include “Start,” “1°,” “2°,” “Flowsheet,” “Review,” and “Transfer.” In some embodiments, the secondary selection areas **1212** may include subsections of a primary category. A user may select a primary selection area **1210** and a secondary treatment selection area **1212**. The selected secondary treatment selection area **1216** may be expanded within the dashboard.

(109) FIG. **12E** depicts an illustrative navigation bar according to some embodiments. As shown in FIG. **12E**, the navigation bar may include a home button **1220** configured to navigate to a home screen, a patient name object **1222** and a patient details object **1224**, and a method of trauma object **1226**. The navigation bar may include a primary notification menu that may include an alerts object **1228** that may be selected to display current alerts, a clinical practice guideline object **1230** that may be selected to display active, relevant clinical practice guidelines, and a checklists object **1232** that may be selected to view current checklists. The navigation bar can also include a secondary notification menu that may include a team object **1234** that may be selected to view and/or assign

members to a treatment team, a display object **1236** that may be selected to view and/or manage output displays, a reports object **1238** that may be selected to view and/or generate reports, a fluids object **1240** that may be selected to view and/or add fluid treatments, and a vitals object **1242** that may be selected to view and/or add vital sign values.

(110) FIGS. **20A-20B** depict an illustrative screen **2001** according to some embodiments for viewing and/or adding vital information **2005** and other patient information, such as Arterial Blood Gas (ABG), Glasgow Coma Scale (GCS) **2010**, or the like. FIG. **12F** depicts an illustrative adjunct bar according to some embodiments. The adjunct bar may include trending selection objects **1244** configured to display current and historical trend values, for example, in a graph format. Non-trending selection objects **1246** may also be presented on the adjunct bar configured to depict current values of specific adjunct selection objects (e.g., vitals). In some embodiments, colors may be used to provide information about the values, such as positive and/or desired values being colored green and negative and/or urgent values being colored red.

(111) FIG. **12G** depicts a trend display according to some embodiments. As shown in FIG. **12G**, a trend display may include a trending data section **1248**. In some embodiments, the trending data section **1248** may be presented as an expanded panel from the data entry/viewing area **1206** depicted in FIG. **12C** configured to present current and historical values in a trend graph. The trend display may include various trend fields **1250** that may be configured to streamline the data entry process by placing symbols, decimal points, and skipping between entry fields automatically for rapid entry of values. Non-limiting examples of trend fields **1250** may include pH, PaPO.sub.2, PaO.sub.2, HCo.sub.3, BD, SaO.sub.2, SBCe, and LAC. The trend display, as well as other screens and displays throughout the healthcare information application, may include a contextual calculator **1252** that may present values, operators, symbols, and layouts based on the context in which they are presented and/or associated content being displayed.

(112) FIGS. **13A-13C** depict illustrative clinical practice guideline process displays according to a first embodiment. As shown in FIG. **13A**, a clinical practice guideline process display may be presented with various steps or processes of a particular clinical practice guideline. In some embodiments, selection of an object (for example, “Penetrating Neck Injury” box) may operate to present further information **1302** about the particular object (for example, information regarding the types of neck injuries, typical symptoms, etc.). In some embodiments, the clinical practice guideline process display may include reference imagery **1303** associated with the clinical practice guideline. The clinical practice guideline process display may indicate which dashboard screen, category, or clinical practice guideline page a user was on for the particular clinical practice guideline, for instance, by using a background color coordinating therewith (e.g., a purple background for a Review category screen). A zoom map **1304** may present the various steps of the clinical practice guideline through selectable objects that may be selected to navigate to a particular step or section of the clinical practice guideline. Selection of a decision point **1305** may be configured to navigate to another page of the clinical practice guideline to continue with the clinical practice guideline process. If a patient transfer has occurred during the clinical practice guideline process, a transfer object **1306** may be selected to automatically generate a timestamp when the transfer occurred. As shown in FIGS. **13A** and **13C**, selection of a treatment object **1307** may generate a timestamp and record the treatment event. FIG. **19** depicts an illustrative screen **1901** depicting a clinical practice guideline **1905** for a gunshot wound according to some embodiments.

(113) FIGS. **14A-14C** depict illustrative clinical practice guideline process displays according to a second embodiment. In the clinical practice guideline process displays depicted in FIGS. **14A-14C**, colors and icons may change for each category of the guideline. In some embodiments, the clinical practice guideline process displays may be configured to use color only for important and/or urgent decisions and/or actions (e.g., decisions **1305** or transfers **1306**).

(114) FIGS. **15A-15E** depict various illustrative GUI screens that may be included in the health

information application in addition and/or corresponding to screens, windows, GUIs, or the like described hereinabove. FIG. 15A depicts an illustrative archive screen **1501** according to some embodiments. The archive screen **1501** may be configured to present archive or historical information relating to patients and/or patient treatment.

(115) In some embodiments, when a user logs into or otherwise access a system screen, the user may be presented with a navigation object, such as navigation object **1225**. In some embodiments, the navigation object **1225** may include a set of selection areas **1245** configured based on default settings, user preference, historical information, location of computing device (for instance, a different set of selection areas may be used if the computing device is located in an emergency room, while another set of selection areas may be used if the computing device is located in an operating room). In some embodiments, the selection areas may include, without limitation, a “Welcome & Support” selection area, a “Team” selection area, an “Add Patient” selection area, an emergency department (ED) selection area, a “Census” selection area, and/or an “Archive” selection area.

(116) In reference to FIG. 12A, FIG. 15A depicts a navigation object **1225** with an active archive selection area **1245**. Selection of the archive selection area **1245** causes an archive information object **1265** to be presented on the archive screen **1501** with various information elements **1275** displayed thereon. FIG. 15B depicts an illustrative census screen according to some embodiments. The census screen **1502** may be configured to provide information about patients receiving treatment at an entity, such as a healthcare facility and/or ER thereof. As shown in FIG. 15B, selection of the census selection area **1245** on the navigation object **1215** causes a census information object **1265** to be presented on the census screen **1502** with various information elements **1275** displayed thereon. In some embodiments, the census platform may include a dynamic document that is filled directly from the clinical documents (healthcare information) that may be configured to keep track of injuries and other issues and to facilitate seamless handover of the patient between treatment teams.

(117) FIG. 15C depicts an illustrative emergency department screen according to some embodiments. The emergency department screen **1503** may be configured to provide information about the emergency department of an entity, such as patients being treated, medical staff and medical professionals, beds, wait times, or the like. In some embodiments, once patients are admitted in the emergency department via the system, their record may be transferred or otherwise associated with the Census platform (see FIGS. 23A-23E).

(118) FIG. 15D depicts an illustrative team and roles screen according to some embodiments. The team and roles screen **1504** may be configured to provide information about a team of medical professionals and staff presently and/or previously treating patients and/or providing administrative or other support. Each team member may be listed along with their role and any other relevant information, such as contact information, expertise, education, experience, or the like. The team members may be filtered **1505** using various criteria, such as by area of expertise (e.g., hand surgery), category (e.g., physicians, nurses, or the like), availability (e.g., team members on-site, call schedules, shift assignments, or the like), or the like. A user may select team members objects **1509** to obtain further information and/or to designate members for a team. In some embodiments, selected team member objects **1509** may be highlighted, for instance, using different colors, font, background, or the like to make the team members more prominent for efficient recognition of an assembled team.

(119) FIG. 15E and FIG. 22 depict an illustrative trauma representation screen according to some embodiments. The trauma representation screen **1506** may include a patient representation **1507** (“trauma man”) that may be configured to allow injuries or other traumas to be directly represented on a virtual patient. In some embodiments, various portions or segments of the patient representation **1507** may be selected and injuries and characteristics thereof specified for the particular area. For instance, a user may select the chest of the patient representation **1507**, a

designated portion thereof, and or some other selection object and input an injury associated with the chest and other information associated therewith (for instance, the severity of the injury, details of the injury, or the like). In some embodiments, selection of the chest of the patient representation **1507**, the designated portion thereof, and or the some other selection object may present information previously entered regarding the injury associated with the chest. In this manner, information may be input using the patient representation **1507** and/or some other portion of the trauma representation screen **1506** and may be retrieved for viewing at a later time. In some embodiments, the symptoms and signs of the patient representation **1507** may be a trigger for generating or making accessible alerts, clinical practice guidelines, and checklists.

(120) As shown in FIG. **22**, the patient representation **1507** may include various information objects **2202a**, **2202b**, for example, relating to injuries, patient characteristics, trauma event characteristics (for instance, was a vehicle accident patient wearing a seatbelt **2202b**). In some embodiments, the information objects **2202a**, **2202b** may be selectable. For example, selection of the airway information object **220a** may cause an airway region **2220** to appear on the trauma representation screen **1506** to allow a medical professional to view and modify information relating thereto. For instance, the airway region **2220** may include airway assessment **2205** information objects and interventions **2210** information objects.

(121) FIGS. **16A** and **16B** depict illustrative review category screens according to some embodiments. FIG. **16A** depicts a review screen **1601** with a summary and plan display (or “wedge”) **1605** (for instance, as an information object **1265**) that may include summaries and/or plans for various portions of the human body. The review screen **1601** may include elements **1606** for specifying certain details regarding the patient and/or treatment thereof. In some embodiments, elements **1606** may be selectable to cause information relating to a particular element (for instance, the face or neck) to be presented on the screen **1601**. In reference to FIG. **12A**, FIG. **16A** depicts a navigation object **1215** with a primary navigation level **1225** and a secondary navigation level **1235**. The selection of the review selection area **1245** causes the selection areas **1255** associated with a review process to be displayed on the secondary navigation level **1235**. Activation of the summary & plan selection area **1255** causes a summary & plan information object **1265** to be presented on the review screen **1601** with various information elements **1275** displayed thereon. FIG. **16B** depicts a review screen **1601** with an associated trend display **1607** (for instance, as depicted in FIG. **12G**).

(122) FIGS. **17A-17E** depict illustrative start category screens according to some embodiments. As shown in FIG. **17A**, a start screen **1701** may include a treatments on scene (TOS) display or wedge **1705**. The TOS wedge **1705** may display a trauma team **1706** for treating a patient. In reference to FIG. **12A**, FIG. **17A** depicts a navigation object **1215** with a primary navigation level **1225** and a secondary navigation level **1235**. The selection of the start selection area **1245** causes the selection areas **1255** associated with a start process to be displayed on the secondary navigation level **1235**. Activation of the treatments on scene selection area **1255** causes a treatments on scene object **1265** to be presented on the start screen **1701** with various information elements **1275** displayed thereon.

(123) As shown in FIG. **17B**, the TOS wedge **1705** may present patient's vitals information **1707**, such as breathing type and oxygen saturation, or other related information (for example, allergies). The TOS wedge **1705** may provide for the viewing and/or entry of data. FIG. **17C** depicts a TOS wedge **1705** that includes a checklist **1708**, such as an adjuncts checklist. As shown in FIG. **17D**, a TOS wedge **1705** may be configured to present fluid information **1709** for a patient. At least a portion of the fluid information **1709** may show the total fluids on a graph using individual boxes that indicate the time and the amount of fluid given at each fluid delivery event. FIG. **17E** depicts a TOS wedge **1705** presenting a data entry screen **1710** for entering the vitals of a patient.

(124) FIG. **18A** depicts an illustrative screen **1801** that includes key patient indicators (KPIs) **1805a-n**. In some embodiments, key patient indicators **1805a-n** may include critical information elements regarding a patient, a trauma related to the patient, the physiological state of the patient,

or the like. In some embodiments, the key patient indicators **1805a-n** may include issues determined about a patient as the patient is going through an assessment, for example, by an emergency medical technician (EMT) team, and emergency room team, a surgical team, or the like. In some embodiments, the key patient indicators **1805a-n** may be presented on the screen **1801** responsive to a user selection, such as a particular navigation selection area, a key patient indicator display object, or the like. In some embodiments, the key patient indicators **1805a-n** may be presented on the screen **1801** by default if a key patient indicator exists. In some embodiments, the key patient indicators **1805a-n** may be presented on the screen **1801** automatically due to a change in the key patient indicators, such as the addition of a new key patient indicator, a change in a key patient indicator (for instance, above or below a predetermined threshold), or the like. In some embodiments, an information element may be designated as a key patient indicator **1805a-n** by a user. In some embodiments, the key patient indicators **1805a-n** may be determined based on an analysis of the patient information, for example, based on historical information regarding key patient indicators used for other patients with similar physiological conditions, demographics, and/or medical conditions. In some embodiments, the key patient indicators **1805a-n** may be determined based on an analysis of the patient information against a set of factors used to determine key patient indicators, for instance, a particular medical condition (for instance, broken vertebrae) may be a key patient indicator for a patient having a certain medical history and/or demographic information (for example, patients 65 years or older), but not for other patients with a different medical history and/or demographic information.

(125) In some embodiments, the healthcare information application is able to generate or designate key patient indicators **1805a-n** by recognizing and tagging key data elements, including combinations of data elements, that experienced clinicians often use to identify dangerous life threatening situations. The key patient indicators **1805a-n** can be used by trauma teams as alerts about the potential severity of an injury, for instance, as a shorthand to communicate important issues quickly and efficiently and/or as triggers for immediate action. FIG. **18B** lists illustrative and non-restrictive key patient indicators according to some embodiments.

(126) FIG. **21** depicts an illustrative screen **2101** for accessing various portions of the body of a patient via a graphical representation **2110** thereof. As shown in FIG. **21**, selection of a navigation object **2105**, such as a navigation object for the abdomen, may cause the healthcare application to present a graphical representation **2110** of the corresponding portion of the body of the patient. In some embodiments, certain areas of the graphical representation **2110** may be selectable to input and/or access information relating thereto. For instance, a user may select the right lumbar region of the graphical representation to view healthcare information corresponding thereto.

(127) FIG. **23A** depicts an illustrative GUI platform according to some embodiments. Multiple healthcare organizations, such as the World Health Organization, have identified timely and complete handover of a patient and patient information as a critical process in optimal patient care. In some embodiments, the census GUI platform may be configured as a mobile, electronic platform that healthcare professionals, such as physicians, may use to track their patients' progress in a healthcare facility, such as a hospital, and seamlessly hand over patient care to their colleagues. The census GUI platform may be configured to apply a highly intuitive user interface and real time analytics to support and enhance networks of communication and collaboration between teams in high intensity, multidisciplinary health systems.

(128) As shown in FIG. **23A**, a GUI may include a navigation object **2305** having a primary navigation level **2325**. Selection of a census selection area **2345** may cause a patient information object **2365** to be presented on the GUI. In some embodiments, the patient information object **2365** may be configured as a patient list, such as a list of patients in a healthcare facility, a department (for instance, an emergency room), affiliated with a particular healthcare professional, combinations thereof, or the like. The patient information object **2365** may include various information elements, including, without limitation, an area where the patient is located **2375a** (for

instance, the emergency room, the operating room, a particular room, or the like), a patient identifier **2375b**, and other patient information **2375c**.

(129) In some embodiments, selection of a patient information element **2375b** may cause the presentation of an illustrative census GUI platform interface according to some embodiments as depicted in FIG. **23B**, for example, configured to allow for access and control of information related to the patient. The GUI may include a navigation object **2305** that includes a primary navigation level **2325** and a secondary navigation level **2335**. Selection of a demographics/patient history selection area **2345** may cause a patient medical history selection area **2355a** and a demographics selection area **2355b** to be presented on the secondary navigation level **2335**. Selection of the demographics selection area **2355a** may cause a demographics information object **2365** to be presented along with various demographics information elements **2375a**. In addition, the GUI depicted in FIG. **23B** may include a vitals information object **2365b** with corresponding vitals information elements **2376b** and a patient data information object **2365c** along with corresponding patient data information elements **2375c**. As shown in FIG. **23C**, selection of the patient medical history selection area **2355a** may cause a medical history information object **2365d** with corresponding medical history information elements **2375d** to be displayed on the GUI.

(130) As shown in FIG. **23D**, selection of the add (or admit) patient selection area **2345** depicted in FIG. **23A** may cause an add patient GUI to be presented on a display element of a client computing device. In some embodiments, adding a patient may generate a new patient record. As shown in FIG. **23D**, a timer **2340** may be depicted, for instance, on the navigation object **2305**. In some embodiments, the timer **2340** may be configured to indicate the time elapsed since a particular event, such as a trauma to a patient, admission, countdown to surgery, or the like. In some embodiments, the duration and/or time scale of the timer **2340** may be determined based on the active selection area **2345** and/or **2355**. In some embodiments, the timer **2340** may be based on phases, for instance, of a process. In some embodiments, the scale of the timer **2340** may be seconds, minutes, days, or some combination thereof. FIG. **23E** depicts a screen presented responsive to selection of the diagnostics selection area **2345**, for example, as depicted in FIG. **23B**. As shown in FIG. **23E**, selection areas **2355** may be presented on the secondary navigation level for various types of diagnostic tests. Selection of a diagnostic test selection area **2355** may cause a diagnostic test information object **2365** and corresponding information elements **2375** to be presented on the screen.

(131) FIG. **24** depicts an illustrative screen template according to some embodiments. Screens and GUI objects presented via the system according to some embodiments may be generated according to the template screen **2401** depicted in FIG. **24**. For example, a screen may include a navigation object **2403** configured according to some embodiments, such as the “circle of life” configuration. The “circle of life” configuration, for example, in a trauma situation, provides a technological advantage as it corresponds to the clock-like progression or flow through the patient treatment process. A patient graphical representation (or “trauma man,” “trauma avatar,” or the like) object **2415** may be presented on the screen. In some embodiments, selection of the patient graphical representation object **2415** may cause a screen, such as the screen **1501** depicted in FIG. **15E**, to be presented to a user with the patient representation **1507** GUI object. A section of the screen **2401** may include immediate alerts **2405** relating to the patient. In some embodiments, the immediate alerts **2405** may include key patient indicators, such as key patient indicators **1805a-n** depicted in FIG. **18A**. A portion of the screen **2401** may include patient demographic information objects **2409** with corresponding patient demographic information elements for a subject patient. The screen **2401** may also include an alerts, clinical practice guidelines, and checklists area **2411**, a data entry/data access area **2413**, an adjuncts to primary survey area **2417**, and a vitals, Arterial Blood Gas (ABG), Glasgow Coma Scale (GCS) area **2419** configured according to some embodiments.

(132) FIG. **15F** depicts an illustrative patient information screen according to some embodiments. The patient information screen **1511** may be configured to present various patient conditions, such

as a shock condition (for instance, a “shock dashboard”). The patient information screen may include selection objects **1515**, selection of which may cause associated information objects **1517a**, **1517b** to be displayed on the patient information screen. In some embodiments, the patient information screen **1511** may include a timeline information object **1513** that may be configured to present a timeline of patient activity, as well as certain events or conditions **1519a-n** that have occurred.

(133) FIG. **25** depicts illustrative and non-restrictive technological advantages resulting from use of the system configured according to some embodiments.

(134) FIG. **26** depicts various symbols and icons that may be used within the healthcare information application to represent information, such as navigation objects, injury icons, or the like. In some embodiments, a user may select an icon for placement on a screen and/or a portion thereof, such as the patient representation **1507** to indicate the location and type of injury.

(135) FIG. **27** depicts a block diagram of exemplary internal hardware that may be used to contain or implement the various computer processes and systems as discussed above. A bus **2700** serves as the main information highway interconnecting the other illustrated components of the hardware. CPU **2705** is the central processing unit of the system, performing calculations and logic operations required to execute a program. CPU **2705** is an exemplary processing device, computing device or processor as such terms are used within this disclosure. Read only memory (ROM) **2730** and random access memory (RAM) **2735** constitute exemplary memory devices.

(136) A controller **2720** interfaces with one or more optional memory devices **2725** to the system bus **2700**. These memory devices **2725** may include, for example, an external or internal DVD drive, a CD ROM drive, a hard drive, flash memory, a USB drive or the like. As indicated previously, these various drives and controllers are optional devices. Additionally, the memory devices **2725** may be configured to include individual files for storing any software modules or instructions, auxiliary data, common files for storing groups of results or auxiliary, or one or more databases for storing the result information, auxiliary data, and related information as discussed above. For example, the memory devices **2725** may be configured to store healthcare information **325**, healthcare analysis processes **330** and/or data contained in the data stores **115**.

(137) Program instructions, software or interactive modules for performing any of the functional steps associated with the analysis and presentation of healthcare information as described above may be stored in the ROM (read only memory) **2730** and/or the RAM (random access memory) **2735**. Optionally, the program instructions may be stored on a tangible computer-readable medium such as a compact disk, a digital disk, flash memory, a memory card, a USB drive, an optical disc storage medium, such as a Blu-ray™ disc, and/or other recording medium.

(138) An optional display interface **2730** may permit information from the bus **2700** to be displayed on the display **2735** in audio, visual, graphic or alphanumeric format. The information may include information related to a current job ticket and associated tasks. Communication with external devices may occur using various communication ports **2740**. An exemplary communication port **2740** may be attached to a communications network, such as the Internet or a local area network.

(139) The hardware may also include an interface **2745** which allows for receipt of data from input devices such as a keyboard **2750** or other input device **2755** such as a mouse, a joystick, a touch screen, a remote control, a pointing device, a video input device and/or an audio input device.

(140) Computer program logic implementing all or part of the functionality previously described herein may be embodied in various forms, including, but in no way limited to, a source code form, a computer executable form, and various intermediate forms (for example, forms generated by an assembler, compiler, linker, or locator). Source code may include a series of computer program instructions implemented in any of various programming languages (e.g., an object code, an assembly language, or a high-level language such as Fortran, C, C++, JAVA, or HTML) for use with various operating systems or operating environments. The source code may define and use various data structures and communication messages. The source code may be in a computer

executable form (e.g., via an interpreter), or the source code may be converted (e.g., via a translator, assembler, or compiler) into a computer executable form.

(141) The computer program may be fixed in a non-transitory form (for example, a source code form, a computer executable form, an intermediate form, or combinations thereof) in a tangible storage medium, such as a semiconductor memory device (e.g., a RAM, ROM, PROM, EEPROM, or Flash-Programmable RAM), a magnetic memory device (e.g., a diskette or fixed disk), an optical memory device (e.g., a CD-ROM), a PC card (e.g., PCMCIA card), or other memory device. The computer program may be fixed in any form in a signal that is transmittable to a computer using any of various communication technologies, including, but in no way limited to, analog technologies, digital technologies, optical technologies, wireless technologies (e.g., Bluetooth), networking technologies, and internetworking technologies. The computer program may be distributed in any form as a removable storage medium with accompanying printed or electronic documentation (e.g., shrink-wrapped software), preloaded with a computer system (e.g., on system ROM or fixed disk), or distributed from a server or electronic bulletin board over the communication system (e.g., the Internet or World Wide Web).

(142) Hardware logic (including programmable logic for use with a programmable logic device) implementing all or part of the functionality previously described herein may be designed using traditional manual methods, or may be designed, captured, simulated, or documented electronically using various tools, such as Computer Aided Design (CAD), a hardware description language (e.g., VHDL or AHDL), or a PLD programming language (e.g., PALASM, ABEL, or CUPL).

(143) It will be appreciated that various of the above-disclosed and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. It will also be appreciated that various presently unforeseen or unanticipated alternatives, modifications, variations or improvements therein may be subsequently made by those skilled in the art which alternatives, variations and improvements are also intended to be encompassed by some embodiments described herein.

Claims

1. A healthcare information analysis and presentation system comprising: a processor; and a non-transitory, computer-readable storage medium in operable communication with the processor, wherein the computer-readable storage medium contains one or more programming instructions that, when executed, cause the processor to: receive point-of-care healthcare information associated with a patient in substantially real-time collected via at least one of a plurality of mobile computing devices; analyze the healthcare information to generate a patient profile of the patient, the patient profile comprising a physiological status, a physiological assessment, and a treatment assessment; store the patient profile in a centralized storage; generate at least one interactive graphical user interface element associated with the patient profile for presentation at a plurality of display devices associated with healthcare professionals involved with care of the patient, wherein the at least one interactive graphical user interface element allows the healthcare professionals to access physiological status, physiological assessment, and/or treatment assessment information from the patient profile via the centralized storage and to provide documentation information and communication information for consumption by at least one other healthcare professional regarding the care of the patient; update the patient profile in the centralized storage with the provided documentation and communication information to thereby provide selective access to the provided documentation and communication information from the patient profile via the centralized storage; and automatically update information displayed on at least one display device of a healthcare professional in substantially real-time based on the updated patient profile to provide up-to-date and dynamic information to the healthcare professional, wherein the updated information displayed on the at least one display device includes a graphical representation of a body of the patient and

- further includes at least one interactive icon depicted on each of a number of body parts for which information is available from the centralized storage, wherein each interactive icon includes a graphical representation of a type of injury associated with the body part and is selectively included in the information displayed on the at least one display device corresponding to a severity of the injury to the patient individually.
2. The system of claim 1, wherein the plurality of display devices comprise a monitor device.
 3. The system of claim 1, wherein the healthcare information comprises at least one of surgeries, symptoms, type of injury, severity of injury, mechanism of trauma, and trauma location.
 4. The system of claim 1, wherein the system is configured to be used for trauma healthcare services.
 5. The system of claim 1, wherein the system is configured to be used for surgical healthcare services.
 6. The system of claim 1, wherein the plurality of mobile computing devices comprise a smartphone and a tablet computing device.
 7. The system of claim 1, wherein the healthcare information comprises user input and device input.
 8. The system of claim 1, wherein the graphical user interface element comprises a dashboard.
 9. The system of claim 8, wherein the dashboard is configured to receive health information user input through at least one field.
 10. A computer-implemented method for analyzing and presenting health information, the method comprising, by a processor: receiving point-of-care healthcare information associated with a patient in substantially real-time collected via at least one of a plurality of mobile computing devices; analyzing the healthcare information to generate a patient profile of the patient, the patient profile comprising a physiological status, a physiological assessment, and a treatment assessment; storing the patient profile in a centralized storage; generating at least one interactive graphical user interface element associated with the patient profile for presentation at a plurality of display devices associated with healthcare professionals involved with care of the patient, wherein the at least one interactive graphical user interface element allows the healthcare professionals to access physiological status, physiological assessment, and/or treatment assessment information from the patient profile via the centralized storage and to provide documentation information and communication information for consumption by at least one other healthcare professional regarding the care of the patient; updating the patient profile in the centralized storage with the provided documentation and communication information to thereby provide selective access to the provided documentation and communication information from the patient profile via the centralized storage; and automatically updating information displayed on at least one display device of a healthcare professional in substantially real-time based on the updated patient profile to provide up-to-date and dynamic information to the healthcare professional, wherein the updated information displayed on the at least one display device includes a graphical representation of a body of the patient and further includes at least one interactive icon depicted on each of a number of body parts for which information is available from the centralized storage, wherein each interactive icon includes a graphical representation of a type of injury associated with the body part and is selectively included in the information displayed on the at least one display device corresponding to a severity of the injury to the patient individually.
 11. The method of claim 10, wherein the plurality of display devices comprise a monitor device.
 12. The method of claim 10, wherein the healthcare information comprises at least one of surgeries, symptoms, type of injury, severity of injury, mechanism of trauma, and trauma location.
 13. The method of claim 10, wherein the method is configured to be performed for trauma healthcare services.
 14. The method of claim 10, wherein the system is configured to be used for surgical healthcare services.

15. The method of claim 10, wherein the plurality of mobile computing devices comprise a smartphone and a tablet computing device.
 16. The method of claim 10, wherein the healthcare information comprises user input and device input.
 17. The method of claim 10, wherein the graphical user interface element comprises a dashboard.
 18. The method of claim 17, wherein the dashboard is configured to receive health information user input through at least one field.
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